

Rising Temperatures and Native Tree Resilience in New England

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Executive Summary

Background

The New England region, which includes Vermont, New Hampshire, Maine, Massachusetts, Connecticut, Rhode Island, and northern New York, is witnessing historic warming levels that have resulted in the region already surpassing benchmarks set by third party agencies. The effects of climate change also extend beyond temperature increases, with precipitation and seasonal patterns fluctuating wildly.

Problem Summary

Research shows that the New England region could see temperature increases of up to 7.6°F, with widespread ramifications resulting from this change. The skiing industry — which has long been an economic driver in many parts of New England — is no longer as stable as it used to be, with decreased snowfall and warmer winters shortening skiing seasons. Increased prevalence of droughts, extreme heat days, and high precipitation days are also being caused by climate change, creating a more volatile environment that is less conducive to tree growth. Additionally, this extreme weather can encourage the presence of harmful pests that feed on and kill tree species, as well as increase the prevalence of existing diseases which further endanger native tree species.

Solution Summary

Two solutions are outlined, both of which impact key stakeholders and are able to be feasibly completed within a reasonable timeframe. The first solution involves pre-commercial thinning, a strategy that can help mitigate infections and harmful pests in forests by selectively thinning trees and artificially selecting for the strongest specimens. This strategy has been used in the past, and could be adapted to address issues in the New England region. The second solution is to assess a wide variety of tree species that can be used as new foundation species for adaptive forests. The results of research contained within this study highlight several key findings that must be accounted for when performing this kind of assessment. Primarily, studies that test the adaptivity of tree species must be conducted in real-world environments, as controlled studies introduce confounding variables that muddle results.

Recommendations

Our primary recommendation for NEFF is to increase tree thinning in the 40,000 acres of land that they own. Thinning in areas with eastern hemlock and American beech trees will play a part in mitigating the spread of beech leaf disease and the hemlock woolly adelgid, two major threats to Northeastern forests. This strategy will encourage healthier trees, provide valuable data for addressing future threats, and buy time for more research into replacing declining forests.

Background

State of the Science

Climate change is a pervasive issue across the United States and the world, but the rising temperatures have impacted New England particularly harshly. The New England region, which includes Vermont, New Hampshire, Maine, Massachusetts, Connecticut, Rhode Island, and occasionally northern New York, has already passed the 1.5 °C warming level and has almost reached the 2 °C level (Young and Young, 2021; Young and Young, 2025). The Intergovernmental Panel on Climate Change set the 1.5 degree level as “do not pass,” meaning the New England region has already warmed to dangerous levels (Masson-Delmotte et al., 2018). Alongside rising temperatures, New England is also predicted to experience significant changes in precipitation patterns, notably an increase in total precipitation and a decrease in number of wet days (Picard et al., 2023). This trend is also reflected in seasonal patterns, with spring starting sooner and seasonal differences increasing each year (Brown, 2025). These climatic changes are impacting New England’s forests. While responses vary from species to species, fruit and grain plants were significantly impacted by extreme temperatures during the reproductive stage (Hatfield and Prueger, 2015).

Species like the European beech (*Fagus sylvatica*) and downy oak (*Quercus pubescens*) require winter temperatures to fall below a certain level in order to release their winter buds, and they will be impacted by the warming winters (Didion-Gency et al., 2024). Likewise, the decrease in soil moisture caused by the rising temperatures negates any positive impacts the rise in temperatures would otherwise have on tree species (Didion-Gency et al., 2022; McMillan 2023).

The eastern hemlock (*Tsuga canadensis*) is one of the foundation species for eastern North American forests, and its loss would disrupt ecosystems across New England (Ellison et al., 2005). The eastern hemlock is threatened by the presence of the hemlock woolly adelgid, an insect which currently doesn’t pose a disastrous threat to eastern hemlocks but that is predicted to thrive in the warmer winters New England is facing in the coming years (Orwig et al., 2012).

The American Beech (*Fagus grandifolia*) is another prevalent species in New England, making up a significant portion of the northern hardwood forests (McCaskill and Morin, 2012). It is also projected to increase in importance in coastal areas in coming years (Busby et al., 2009). The American Beech population is currently suffering from the Beech leaf disease, an insect-fungus complex of the beech scale insect (*Cryptococcus fagisuga*) and the beech bark canker fungus (*Neonectria faginata*), plaguing the eastern United States, which is killing many trees and impacting many more (Shepherd et al., 2025; McCaskill and Morin, 2012).

Relevant Policies

Despite New England’s sensitivity to climate change, many states are reconsidering their climate plans, with rising electricity bills for their residents and the federal government’s

hostility towards renewable energy (Plumer, 2026). For example, Rhode Island's governor, Dan McKee, has proposed pushing back the state's 2033 deadline for their 100% Renewable Energy Standard, in hopes to save taxpayers \$572 million by 2031 (Phillips, 2026).

On a national level, the USDA Secretary Rollins announced in June of 2025 that the federal government plans to rescind the 2001 Roadless Area Conservation Rule, which was established under the Clinton administration to prohibit road construction and timber harvesting on inventoried "roadless areas" on National Forest System lands (U.S. Department of Agriculture, 2025; U.S. Department of Agriculture, 2001). Secretary Rollins states that the reason for the rescinding of the rule is both due to the high risk of wildfire in many inventoried forests and to allow for economic and road development in those forests. This has worried many environmental advocates, since roadless areas make up over 70% of the national forest watersheds; in New Hampshire, the Merrimack river is struggling with pollution as it is, but water from the White Mountains helps offset the water quality problems (Delgado et al., 2011; Wisniewski, 2026). Road building could also impact aquatic ecosystems and stream structure, which increases the likelihood of greater flooding. In New England, roadless areas include parts of Robert T. Stafford White Rocks National Recreation Area, Green Mountain and Finger Lakes National Forests, and Green Mountain National Forest in Vermont, and parts of the White Mountain National Forest in New Hampshire (U.S. Forest Service Enterprise Map Services Program, 2001).

American Beech and eastern hemlock are both in the top ten most populous tree species in the Vermont timberlands, which include the Green Mountain National Forest. Eastern hemlocks are one of the dominant trees in New Hampshire's hemlock-hardwood-pine forests, which is the most wide-spread habitat in the state, and American Beech trees are one of the dominant trees in New Hampshire's northern hardwood-conifer forest, which comprises many parts of the White Mountain National Forest (Wharton et al., 2003; University of New Hampshire Cooperative Extension, 2011a; University of New Hampshire Cooperative Extension, 2011b). The potential decrease in roadless areas could lead to an increase in stressors for these crucial trees in New England's forests.

Audience Consideration

Our primary audience for this paper is the New England Forestry Foundation (NEFF), a non-profit organization that owns and manages forests, advocates for forest conservation policies, and advises private landowners on forest management. NEFF created a forest management approach called Exemplary Forestry. This aims to enhance the role of forests in mitigating climate change, improve biodiversity and wildlife habitats, and grow and harvest more sustainably produced wood (New England Forestry Foundation, 2026). The specifics of what this approach entails depends on the forest for which it is tailored. For central and transitional hardwood forests, for example, Exemplary Forestry includes implementing best management practices to protect and improve forest conditions, and implementing advanced silviculture, which itself includes continuously improving forest stands, maintaining connectivity

between habitats, and providing conditions which are well-suited to the specified umbrella wildlife species (Worthley et al., 2022). Implementing advanced silviculture also includes growing tree species well-suited to each site, which accounts for matching to soil conditions and expected changes in climate conditions (Worthley et al., 2022). Our research aims to inform as to which trees are well suited to withstand the changing climate conditions in New England, and thus contribute to NEFF's Exemplary Forestry approach.

Problem Description

Overview of the Problem

The New England region is rapidly warming due to climate change, with temperature increases outpacing those of other regions in the United States (Young and Young, 2025). While most warming has occurred since the 1980s, an acceleration in recent years has been observed, and the effect of this warming is wide reaching. Winter periods are facing the brunt of this change, with minimum and nighttime temperatures increasing and snow cover dramatically decreasing. Consequently, several key New England industries, such as the skiing industry, for example, have been severely impacted (Brown, 2025). As temperatures increase in southern regions, snowfall — and skiing — has shifted northward, reducing income and affecting local communities that once relied upon that tourism traffic for survival.

The degree to which the New England region will warm is still yet to be determined. However, previous research has identified a low-emission and high-emission scenario for the region, with projected temperature increases by the end of the century reaching 2.6 °F and 7.6 °F, respectively (Janowiak et al, 2018). Under the high emission scenario, extreme heat events (days with maximum temperature exceeding 90 °F) could increase by 16 to 60 days per year. Likewise, high rainfall events are projected to become a more common occurrence, underscoring the increase in weather extremes expected due to climate change.

A major effect of this changing climate is the unpredictability of weather in coming years. Young and Young (2021) described how the New England region's four-season climate is slowly being eroded. This not only has ramifications for the winter season, as outlined above, but also for the summer season. Drought is becoming increasingly prevalent, which has enabled tree-harming diseases and pests such as the spongy moth (Brown, 2025). Outbreaks of the spongy moth and other insects like the hemlock woolly adelgid have been more frequent in recent years, and research suggests that the prevalence of these attacks will only increase as climate change progresses (Orwig et al, 2012).

Another disease affecting tree populations in the region is beech leaf disease. Because of *Fagus grandifolia*'s dominance in the northeast, the rise of beech leaf disease is particularly worrisome. A relatively new disease, beech leaf disease has been shown to dramatically decrease

beech tree populations, especially in areas where beech trees are highly concentrated (Shepherd et al, 2025). Beech leaf disease has been shown to be most present when climate factors are stable (Selvi et al, 2026). This coincides with natural distributions of beech trees, as most beech populations lie within a low-precipitation seasonality band.

As seen in Figure 1, a band of consistent precipitation seasonality stretches from lower Appalachia to the Northeast. In Figure 2, beech tree concentration is shown, with a strong overlap of high beech presence existing along the stable band. As such, trees like the American beech could be at risk, as diseases like the beech leaf disease are thriving in the tree's natural habitat. Such situations could push the beech tree northward, with warming summers in southern regions exacerbating this trend.

Figure 1. Precipitation seasonality across the continental U.S. Precipitation seasonality measures the difference in precipitation in low precipitation versus high precipitation seasons. This measure was min-max normalized to a 0-1 scale. Higher values indicate a more volatile environment. The survey covers the eastern half of the United States, with each grid cell measuring 30 minutes of longitude by 30 minutes of latitude. White areas did not have reported data in the survey. (Figure 1 created using data from Record & Ellison, 2023)

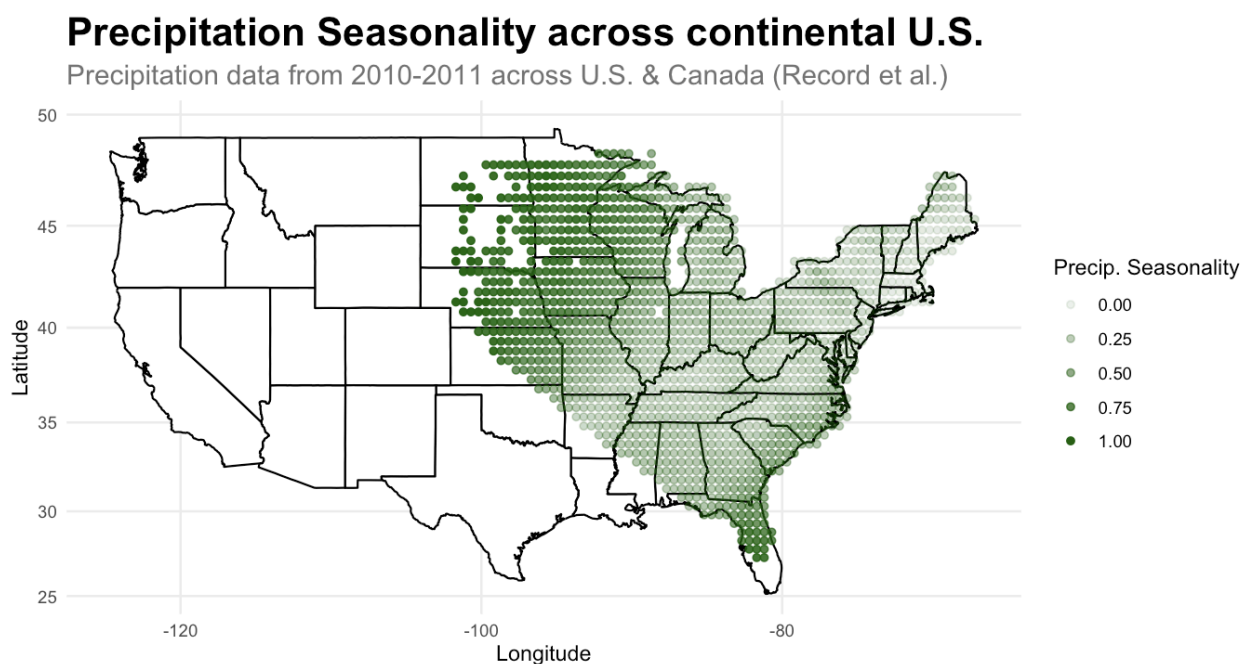
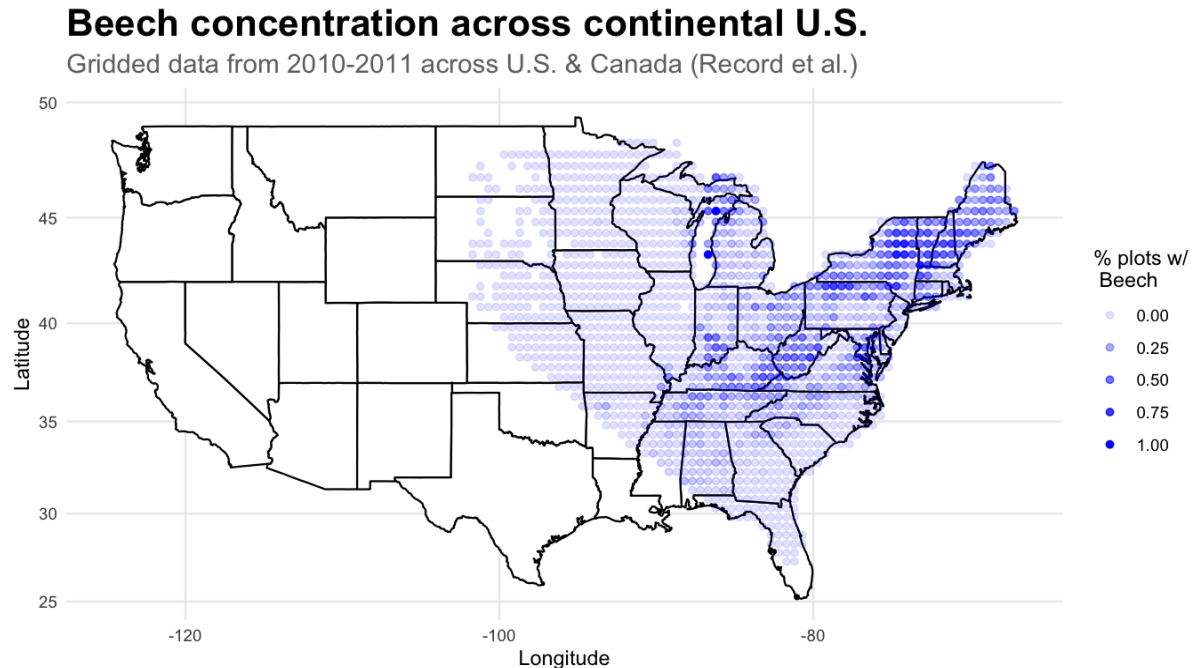


Figure 2. Beech concentration across the continental U.S. Concentration is defined by the percentage of plots in each grid square that reportedly contained Beech trees. The survey covers the eastern half of the United States with each grid cell measuring 30 minutes of longitude by 30 minutes of latitude. White areas did not have reported data in the survey. (Figure 2 created using data from Record & Ellison, 2023)



Stakeholder Values

Overall, several challenges face trees in the New England region. Firstly, warming climates are directly impacting tree populations and ecosystems. Second, warming climates are enabling insects that threaten tree species. Lastly, native tree diseases are continuing to spread, and the intersection between climate factors and already present diseases could spell disaster for old growth forests that have been around for centuries. This dramatic threat for native tree species is not only consequential for residents of the New England region, but also for forest conservationists, reforestation experts, and researchers. Conservationists interested in preventing the decrease of these old-growth forests will be facing major challenges. Likewise, reforestation experts will be interested in what trees are most suitable to replace those now-gone forests, and more research is needed in this area. Finally, researchers will look to identify the factors that are causing tree decline as well as why certain species are most suited to survive in a warming climate. Each of these factors requires adequate monitoring and extensive research, especially as the climate continues to rapidly change in this region.

Doing nothing — failing to perform tests, survey tree counts, identify new foundation species, and assess threats — will result in a society unequipped to deal with a foreign climate. Local industries that drive revenue in New England, such as skiing and timber, will continue to fade, causing economic uncertainty for residents. Conservationists will find rapidly diminishing

forests and reduced environmental stability. Reforestation experts and researchers will have fewer samples to study, will have an incomplete understanding of what is driving forest decline, and will be unable to replace these trees effectively. Overall, as forests disappear and ecosystems form anew, a lack of understanding regarding what is occurring and why it's occurring could be catastrophic. Conversely, by implementing the above methods, the New England region will be more prepared for weathering the coming storm, both literally and metaphorically.

Solution Description

Acceptable Solution Criteria

Acceptable solutions must include a plan for widespread implementation that will improve the health of forests across New England in the short and long term. They must also be self-contained, meaning that only the New England Forestry Foundation needs to participate for the solution to be thoroughly effective in at least parts of New England. For example, reducing greenhouse gas emissions in New England to slow climate change is an admirable goal, but without the rest of the world's participation, the reduction would not be sufficient to affect the health of New England forests in a meaningful way.

The nature of climate-related goals is such that many targets are frequently abandoned or not reported on after the initial announcement. A study into corporate emissions goals found that nearly 40% failed to achieve the goal or stopped reporting on it (Jiang et al, 2025). Thus, the need for accountability and feasibility of completion is integral to solution success. Additionally, without a direct, grounded link to the communities that the solution will be impacting, any such plan will be either doomed to fail or create adverse consequences. These factors are thus integral for evaluating potential solutions, and the proposed solutions below are ones that match the above criteria.

Solution #1: Expand Pre-Commercial Thinning Efforts

Solution Overview

Pre-commercial thinning (PCT) is a forest management method that involves cutting down a certain portion of trees in a stand to decrease tree density and help the growth of the remaining vegetation (Regan-Loomis, 2025). In the past, the process has been used by timber harvesters so they can plant the most trees possible and artificially select for the strongest, healthiest ones later, as opposed to depending on the growth and health of fewer trees (Emmingham et al, 1984). However, it can also be used as a method to help forests adapt to the climate through artificial selection. Foresters can remove trees that are suffering from heat stress, water stress, or climate-related disease outbreaks, leaving the more resilient trees more resources

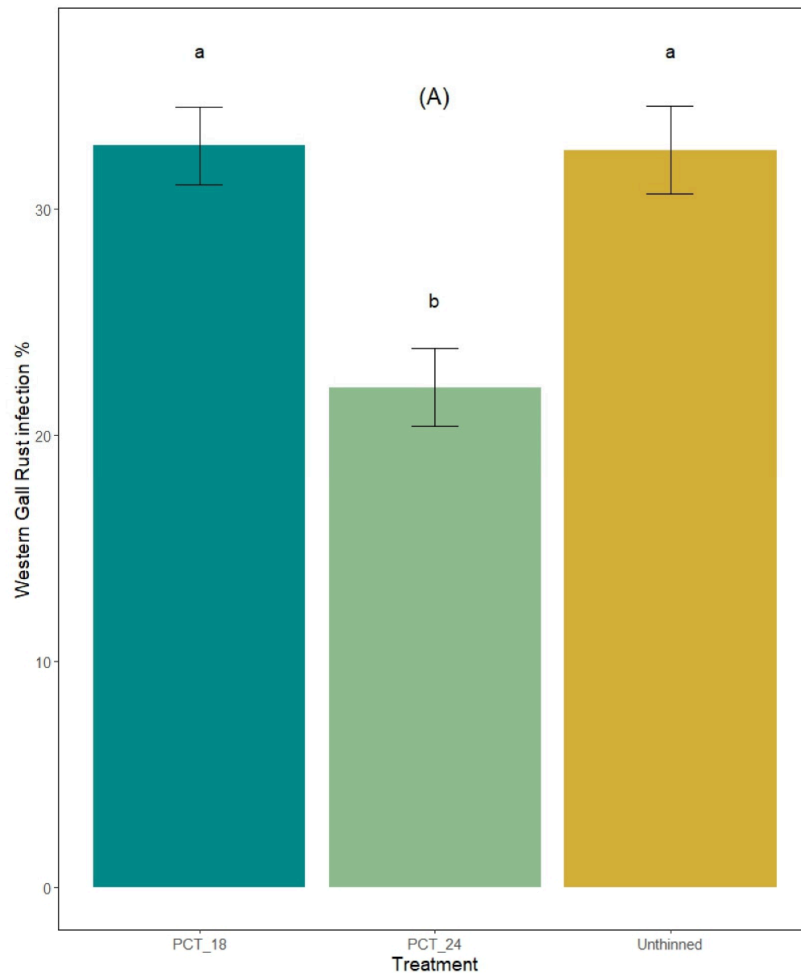
to grow and thrive. Those trees will then be the ones that produce seeds, and new trees that grow will hopefully have the genes that caused their predecessors' resilience.

Some forests in New England have begun pre-commercial thinning as a method of sustainable forestry already. Recently, Great Mountain Forest (GMF) in Connecticut has begun using the practice to attempt to conserve their forest by prioritizing the growth of trees that have been most resilient to recent climate changes and disease outbreaks (Regan-Loomis, 2025). While they specify that they are attempting to preserve the diversity of the ecosystem and avoid a monoculture, the reality of the situation means that if disease outbreaks are concentrated on certain species, those species will be thinned at a larger scale than others. In the Great Mountain Forest, this is demonstrated most strongly with beech trees, which have been under attack by beech leaf disease for the last decade. According to Elizabeth Ward at the Connecticut Agricultural Experiment Station (CAES), "In Connecticut, every single beech tree has beech leaf disease," (Linden 2026). Most beech trees are thus being removed from GMF stands. This is a downside of pre-commercial thinning: while it aids forests in adapting to a changing climate and changing ecosystems, it is not a solution to the root causes of these shifts, preventing the death of the entire forest while risking irreversible changes to the ecosystem.

There is, however, hope that pre-commercial thinning could help a small amount with tree infections or infestations. A study in Alberta on lodgepole pines found that the proportion of trees with western gall rust infections decreased by over 10% compared to the control group in stands whose trees were thinned between 23-25 years of age (Figure 3) (Scaria & Pinno, 2024). The authors suggest this is because the chances of western gall rust infection decrease greatly after a tree reaches 20 years of age. A few years after 20, most of the trees susceptible to infection are already infected, so after removing the infected trees, the remaining trees are less likely to be infected.

A direct consequence that one might think of when faced with a plan that involves cutting down trees is the loss of carbon sequestration potential. However, a study by Schaedel et al. (2017) suggests this might not be an issue. The study was done on stands of western larch, which over 54 years were thinned either 0 (control), 1, 2, or 4 times. When researchers measured the carbon stores, they found that the amount of thinning caused no significant variance in the amount of stored carbon in the stand. This is likely because thinning promotes the growth of vegetation that would normally be blocked from sunlight by dense treetops, as well as allowing the remaining trees to grow bigger. Both of those increase the stand's carbon stores, and this effect appears to make up for the fewer number of trees. This is good news for PCT efforts in New England because conservationists would not have to worry about a trade-off between mitigating greenhouse gas emissions and helping their local forests adapt to the changing climate.

Figure 3. Percentage of western gall rust infection in lodgepole pine stands thinned at different life stages. PCT_18 refers to tree stands that were thinned at 17-19 years of age and PCT_24 refers to tree stands that were thinned at 23-25 years of age. (Figure 3 from Scaria & Pinno, 2024)



Stakeholder Impacts

The New England Forestry Foundation has had some experience with pre-commercial thinning already, but all previous projects have been small-scale compared to the amount of land they own or help conserve (McGuire, 2024). This is beneficial because they already have an internal model of how to implement these practices – they just have to implement them at scale. In addition, NEFF is a nationally recognized conservation organization, even receiving a \$32 million grant from the U.S. Department of Agriculture in 2024 (McGuire, 2024). If NEFF implements PCT on their lands, it could serve as a model and catalyst for other landowners and government environmental agencies to do the same. The timber industry may be able to be persuaded to expand PCT implementation as well, since it's also a technique to increase timber yield (Emmingham et al, 1984). All these entities would be working towards healthy, diverse

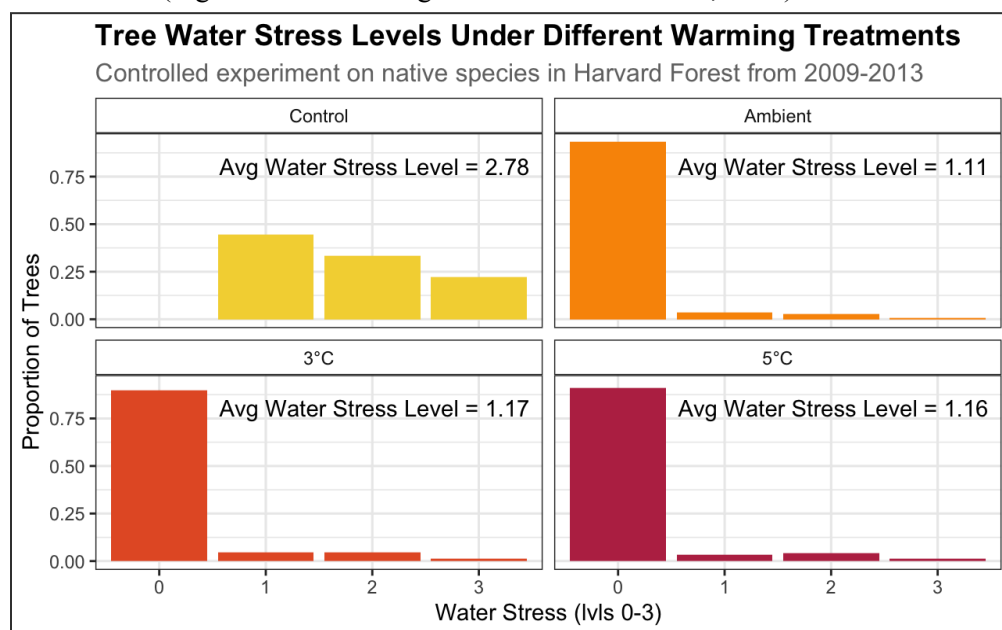
forests despite the changing climate, preserving the natural beauty and crucial ecosystems of New England.

Solution #2: Widespread Assessment to Find Resilient Tree Species

Solution Overview

The loss of a foundation species in a forest ecosystem can be cataclysmic (Ellison et al, 2025). As trees like the eastern hemlock face increasing challenges, new species will need to be identified that are resistant to these threats. Several factors must be taken into account when identifying those species. Firstly, research into the effects of certain factors on tree growth must be completed in real-world environments. The result of our research shows that controlled in-chamber experiments failed to capture many of the climate effects present in the real world, and may have introduced artificial stability that benefitted tree growth even when conditions were considered “extreme.” For example, the “water stress” variable present in Melillo et al. (2023) was set to zero in over 90% of cases for all of the in-chamber trees, while the control group — which was exposed to the outside environment — showed no zero scores and nearly a quarter of control trees were assigned the highest water stress score.

Figure 4. Average tree water stress levels under different warming treatments. Non-control groups of trees were kept in closed chambers that were heated to the ambient outside temperature, 3°C above ambient, and 5°C above ambient, respectively. The lowest water stress level (no stress) is 0 and the highest is 3. The species studied were *Acer barbatum*, *Acer rubrum*, *Acer saccharum*, *Betula alleghaniensis*, *Betula lenta*, *Betula populifolia*, *Betula papyrifera*, *Carya glabra*, *Fagus grandifolia*, *Fraxinus americana*, *Ilex vomitoria*, *Liquidambar styraciflua*, *Liriodendron tulipifera*, *Magnolia grandiflora*, *Magnolia virginiana*, *Nyssa sylvatica*, *Pinus palustris*, *Pinus resinosa*, *Pinus strobus*, *Pinus taeda*, *Quercus alba*, *Quercus falcata*, *Quercus nigra*, *Quercus phellos*, *Quercus rubra*, *Quercus velutina*, and *Ulmus americana*. (Figure 4 created using data from Melillo et al., 2023)



This discrepancy highlights a key difference between controlled experiments and real world ecology. Trees that are tested in closed environments are not being evaluated for their suitability to different climate scenarios, they are being evaluated for their suitability to different labs. Our analysis of the differences between Record and Ellison (2023) and Melillo et al. (2023) thus suggests that tree species should be thoroughly evaluated through real world methods that avoid artificially introduced confounding variables.

Stakeholder Impacts

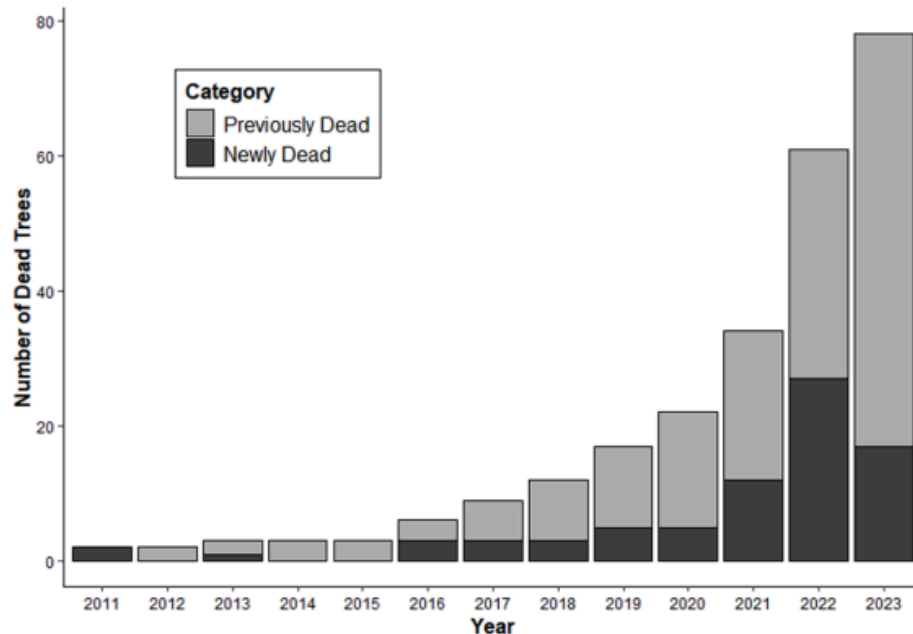
The New England Forestry Foundation could use the results of such an assessment to recommend action plans for coming deforestation as a result of climate change and could also inform NEFF's own efforts to continue ongoing forest conservation initiatives. However, this solution is not all-encompassing and fails to mitigate the ongoing loss of foundation species in the northeast. While the solution is relevant to both researchers and reforestation experts alike, as it provides key data for assessing future tree species that can replace declining populations, it does significantly improve the lives of locals to the region or help preserve already-existing forests.

Recommendations

Our recommendation to the New England Forestry Foundation is to increase thinning efforts to curb the spread of infestations and infections plaguing New England forests. As discussed above, both the American beech and the eastern hemlock are struggling with external attacks which promise to only get worse with the rapidly rising temperatures. We advise an increase in thinning of trees in highly infested areas to limit the spread and impact of the hemlock woolly adelgid and the beech leaf disease. As shown in Figure 5, the total amount of trees killed by beech leaf disease is constantly increasing, and the number of trees killed in a given year is also, on the whole, increasing.

Given that NEFF both manages their own forests and assists private landowners in managing their land, we prompt NEFF to increase tree thinning initiatives for NEFF-owned forests, to then model and recommend for private landowners. Since NEFF isn't part of the federal government, it can take more immediate action to prevent the spread of disease and infestation in the 40,000 acres of NEFF-owned land and the 1.2 million acres of land NEFF has helped conserve (New England Forestry Foundation, 2024b). We urge NEFF to begin pre-commercial thinning as soon as possible, as the New England climate will only become more and more hostile to fragile species over time. While the hemlock woolly adelgid doesn't currently pose as large of a threat as the beech leaf disease, its impact will only increase with the rising temperatures, meaning that if it is not curbed now, it may significantly impact one of New England's most important tree species.

Figure 5. Cumulative number of dead American beech trees in Squire Valleevue and Valley Ridge Farm in Hunting Valley, Ohio. Black segments of the bars indicate the number of beech newly dead in that year while the gray segments indicate the trees that died in previous years (beginning in 2011) (Shepherd et al., 2025).



We encourage prioritizing NEFF-owned, public-facing land, to drum up public interest and support as well as develop methods and more detailed recommendations for private landowners.

Not only would tree thinning increase the health and quality of NEFF-owned and managed forests, which would in turn increase public use of forests, but an uptick in forest management would also create more job opportunities to further support New England communities. As part of their mission statement, NEFF is future-focused and aims to keep forests healthy in spite of and to aid in lessening climate change. With the plague of Beech leaf disease currently ravaging forests and the infestation of woolly hemlock adelgids on the verge of a serious outbreak in the coming warm winters, tree thinning can play a role in reducing the active threats to New England forests, which, in turn, supports forest health in the long run.

Conclusions

The health of New England's forests, like many ecosystems around the world, is being compromised by anthropogenic climate change. Warming temperatures can cause an increase in water stress and affect trees' phenology, as well as promote the spread of harmful insects and diseases such as beech leaf disease and the hemlock woolly adelgid. This in turn increases tree mortality, which has reverberating negative effects throughout the ecosystem and New England

as a whole. Unfortunately, several recent state and federal policies aim to halt or even reverse environmental protection measures, suggesting that climate change itself is not going to stop anytime soon. As such, ecosystems like forests must adapt. If conservation organizations like the New England Forestry Foundation do nothing to help forests survive in a changing climate, there would be tragic consequences for not only the ecosystem, but the human industries that rely on forests and every person who enjoys the wonder of nature.

Two possible courses of action for the NEFF were proposed: the expansion of pre-commercial thinning and a widespread assessment of the resiliency of different tree species. Pre-commercial thinning would be a more immediate solution, removing infected or damaged trees to prioritize the more resilient individuals. An assessment would give conservationists a more comprehensive knowledge of how climate change affects different species, informing future conservation strategies.

Of these two solutions, our recommendation for the NEFF is to expand pre-commercial thinning efforts in the forest lands they own, providing direct assistance to forest adaptation. PCT has the potential to decrease the spread of disease among trees, and though it involves tree removal, has not been shown to impact carbon stores. By choosing poorly adapted trees to remove, this practice will artificially select for more resilient trees, improving forest resilience into the future. Since the NEFF is a respected conservation organization, our hope would be that their efforts would serve as an example to other landowners, organizations, and government agencies to collectively work towards New England forests that are healthy, resilient, and adaptable.

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